

Chapter 6

Matrices

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Introduction and Algebra of Matrices]

1. Let A be a 5×8 matrix, then each column of A contains
(1) 5 elements (2) 8 elements (3) 40 elements (4) 13 elements

Sol. Answer (1)

2. If A is a matrix of order 10×15 , then each row of A contains
(1) 25 elements (2) 15 elements (3) 10 elements (4) 150 elements

Sol. Answer (2)

3. The number of all possible matrices of order 2×3 with each entry 1 or -1 is
(1) 32 (2) 12 (3) 6 (4) 64

Sol. Answer (4)

4. If A is of order $m \times n$ and B is of order $p \times q$, then AB is defined only if
(1) $m = q$ (2) $m = p$ (3) $n = p$ (4) $n = q$

Sol. Answer (3)

5. If P is of order 2×3 and Q is of order 3×2 , then PQ is of order
(1) 2×3 (2) 2×2 (3) 3×2 (4) 3×3

Sol. Answer (2)

6. If $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, then

- (1) $A^2 = 0$ (2) $A^2 = A$ (3) $A^3 = A$ (4) $A^2 = 2A$

Sol. Answer (1)

7. If $A = \begin{bmatrix} x & y & z \end{bmatrix}$, $B = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$, $C = [\alpha \ \beta \ \gamma]^T$, then ABC is

- (1) Not defined (2) Is a 3×3 matrix (3) Is a 1×1 matrix (4) Is a 3×2 matrix

Sol. Answer (3)

8. If $A + B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $A - 2B = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$, then A is equal to

- (1) $\frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ (2) $\frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Sol. Answer (1)

9. If $X + \begin{bmatrix} 2 & 1 \\ 6 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then ' X ' is equal to

- (1) $\begin{bmatrix} 0 & 1 \\ 0 & 6 \end{bmatrix}$ (2) $\begin{bmatrix} 0 & -1 \\ 0 & -6 \end{bmatrix}$ (3) $\begin{bmatrix} -1 & 0 \\ -6 & 0 \end{bmatrix}$ (4) $\begin{bmatrix} 1 & 0 \\ 6 & 0 \end{bmatrix}$

Sol. Answer (3)

10. If $A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 5 & 7 \end{bmatrix}$ and $2A - 3B = \begin{bmatrix} 4 & 5 & -9 \\ 1 & 2 & 3 \end{bmatrix}$, then B is equal to

- (1) $\frac{1}{3} \begin{bmatrix} -2 & -1 & 15 \\ 5 & 8 & -11 \end{bmatrix}$ (2) $\begin{bmatrix} 2 & 1 & -15 \\ 5 & -8 & -11 \end{bmatrix}$ (3) $\begin{bmatrix} 2 & -1 & 15 \\ 5 & 8 & 11 \end{bmatrix}$ (4) $-\frac{1}{3} \begin{bmatrix} 2 & 1 & -15 \\ 5 & -8 & -11 \end{bmatrix}$

Sol. Answer (4)

11. If $\begin{bmatrix} x & 1 \\ -1 & -y \end{bmatrix} + \begin{bmatrix} y & 1 \\ 3 & x \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, then

- (1) $x = -1, y = 0$ (2) $x = 1, y = 0$ (3) $x = 0, y = 1$ (4) $x = 1, y = 1$

Sol. Answer (2)

12. Let $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 0 & 2 \\ 3 & 4 & 5 \end{bmatrix}$ and $A + B - 4I = 0$, then B is equal to

- (1) $\begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & -2 \\ -3 & -4 & -1 \end{bmatrix}$ (2) $\begin{bmatrix} 2 & 3 & 5 \\ 1 & -4 & 2 \\ 3 & 4 & 1 \end{bmatrix}$ (3) $\frac{1}{4} \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & -2 \\ -3 & -4 & -1 \end{bmatrix}$ (4) $\begin{bmatrix} 2 & 3 & 5 \\ -1 & 4 & -2 \\ 3 & 4 & 1 \end{bmatrix}$

Sol. Answer (1)

13. If $A = \begin{bmatrix} 1 & 2 \\ -1 & 8 \\ 4 & 9 \end{bmatrix}$ and $X + A = 0$, then X is equal to

(1) $\begin{bmatrix} 1 & 2 \\ 1 & 8 \\ 4 & 9 \end{bmatrix}$

(2) $\begin{bmatrix} -1 & -2 \\ 1 & 8 \\ 4 & 9 \end{bmatrix}$

(3) $\begin{bmatrix} 1 & 2 \\ 1 & -8 \\ 4 & 9 \end{bmatrix}$

(4) $\begin{bmatrix} -1 & -2 \\ 1 & -8 \\ -4 & -9 \end{bmatrix}$

Sol. Answer (4)

14. If $\begin{bmatrix} x+y & y-z \\ z-2x & y-x \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$, then

(1) $x = 2, y = 1, z = 3$

(2) $x = 3, y = 1, z = 2$

(3) $x = 1, y = 2, z = 3$

(4) $x = 1, y = 3, z = 2$

Sol. Answer (3)

15. If $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \end{bmatrix}$. The matrix 'C' such that $A + B + C$ is null matrix, is

(1) $\begin{bmatrix} -3 & 4 & -1 \\ -3 & 0 & -1 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & 3 \end{bmatrix}$

(3) $\begin{bmatrix} -1 & -2 & 3 \\ 1 & 2 & -3 \end{bmatrix}$

(4) $\begin{bmatrix} 3 & 4 & 1 \\ 3 & 0 & 1 \end{bmatrix}$

Sol. Answer (1)

16. If A is a matrix of order $[x-1] \times 3$ and B is a matrix of order $3 \times [y-2]$, where $[]$ represents greatest integer function, such that AB is a matrix of order 4×5 , then

(1) $x \in [5, 6)$

(2) $x \in [5, 6]$

(3) $y \in [7, 8]$

(4) $y \in [8, 9]$

Sol. Answer (1)

$$[x-1] = 4 \Rightarrow 4 \leq x-1 < 5 \Rightarrow 5 \leq x < 6$$

$$[y-2] = 5 \Rightarrow 5 \leq y-2 < 6 \Rightarrow 7 \leq y < 8$$

17. If a diagonal matrix is commutative with every matrix of the same order then it is necessarily

(1) A diagonal matrix with atleast two diagonal elements different

(2) A scalar matrix

(3) A unit matrix

(4) A diagonal matrix with exactly two diagonal elements different

Sol. Answer (2)

Let, $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$, and $B = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{bmatrix}$

$$AB = \begin{bmatrix} ax_1 & ax_2 & ax_3 \\ bx_4 & bx_5 & bx_6 \\ cx_7 & cx_8 & cx_9 \end{bmatrix} \text{ and } BA = \begin{bmatrix} ax_1 & bx_2 & cx_3 \\ ax_4 & bx_5 & cx_6 \\ ax_7 & bx_8 & cx_9 \end{bmatrix}$$

$$AB = BA, \text{ if } a = b = c,$$

$$\text{then, } A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$$

which is a scalar matrix.

18. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, then A^{2008} is equal to

(1) $\begin{bmatrix} 2008 & 0 \\ 1 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & 0 \\ 2008 & 1 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & 0 \\ 1 & 2008 \end{bmatrix}$ (4) $2007 \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

Sol. Answer (2)

19. If for a matrix A , $A^2 + I = 0$, where I is the identity matrix, then A equals

(1) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$ (3) $\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Sol. Answer (2)

20. If A and B are two matrices such that $AB = B$ and $BA = A$, then $A^2 + B^2$ is equal to

(1) $2AB$ (2) $2BA$ (3) AB (4) $A + B$

Sol. Answer (4)

21. $\begin{bmatrix} 7 & 1 & 2 \\ 9 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ is equal to

(1) $\begin{bmatrix} 45 \\ 44 \end{bmatrix}$ (2) $\begin{bmatrix} 43 \\ 45 \end{bmatrix}$ (3) $\begin{bmatrix} 44 \\ 43 \end{bmatrix}$ (4) $\begin{bmatrix} 43 \\ 50 \end{bmatrix}$

Sol. Answer (4)

22. If $f(x) = x^2 + 4x - 5$ and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$, then $f(A)$ is equal to

(1) $\begin{bmatrix} 0 & -4 \\ 8 & 8 \end{bmatrix}$ (2) $\begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (4) $\begin{bmatrix} 8 & 4 \\ 8 & 0 \end{bmatrix}$

Sol. Answer (4)

[Types of Matrices and their Properties]

23. If A is a square matrix such that $A^2 = I$, then A^{-1} is equal to

- (1) I (2) 0 (3) A (4) $I + A$

Sol. Answer (3)

24. $\cos\theta \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \sin\theta \begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$ is equal to

- (1) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (4) $-\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Sol. Answer (2)

25. If the square matrices A and B are such that $AB = A$ and $BA = B$, then

- (1) A is an idempotent matrix but B is not
 (2) B is an idempotent matrix but A is not
 (3) A and B are both idempotent matrices
 (4) Neither A nor B are idempotent matrices

Sol. Answer (3)

Given $AB = A$... (1)

$AB = A$

Pre multiplying by B

$B(AB) = BA$

$(BA)B = B$

$B^2 = B$

Similarly $A^2 = A$

Hence, B is idempotent. Similarly, A is idempotent.

26. Which of the following about the trace of a matrix is false?

- (1) $\text{tr}(ABC) = \text{tr}(BCA) \neq \text{tr}(ACB) = \text{tr}(BAC) = \text{tr}(CBA)$
 (2) $\text{tr}(AB) = \text{tr}(BA)$
 (3) $\text{tr}(A - B) = \text{tr} A - \text{tr} B$
 (4) $\text{tr}(A^2) = (\text{tr} A)^2$

Sol. Answer (4)

Let, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\text{tr}(A) = a + d$

$$A^2 = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}$$

$$\therefore \text{tr}(A^2) = a^2 + 2bc + d^2$$

$$\text{and } (\text{tr}A)^2 = (a + d)^2 = a^2 + 2ad + d^2$$

$$\therefore \text{tr}(A^2) \neq (\text{tr}A)^2$$

27. If A , B and $A + B$ are idempotent matrices, then

$$(1) AB = BA \quad (2) AB + BA = O \quad (3) AB - BA = I \quad (4) AB + BA = I$$

Sol. Answer (2)

$$(A + B)^2 = A^2 + B^2 + AB + BA$$

$$\Rightarrow A + B = A + B + AB + BA$$

Now, $AB + BA = O$ (using cancellation law)

28. If $A = [a \ b]$, $B = [-b \ -a]$ and $C = \begin{bmatrix} a \\ -a \end{bmatrix}$, then correct statement is

$$(1) A = -B \quad (2) A + B = A - B \quad (3) AC = BC \quad (4) CA = CB$$

Sol. Answer (3)

$$B = [-b \ -a]$$

$$\therefore -B = [b \ a] \neq A$$

$$\therefore A + B = [a - b \ b - a] \text{ and } A - B = [a + b \ b + a]$$

$$\Rightarrow A + B \neq B + A$$

$$\text{and } AC = [a \ b] \begin{bmatrix} a \\ -a \end{bmatrix} = [a^2 - ab]$$

$$BC = [-b \ -a] \begin{bmatrix} a \\ -a \end{bmatrix} = [a^2 - ab]$$

29. Let $A = \begin{bmatrix} 1 & x/n \\ -x/n & 1 \end{bmatrix}$, then $\lim_{n \rightarrow \infty} A^n$ is

$$(1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (2) \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} \quad (3) \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix} \quad (4) \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Sol. Answer (2)

$$A = \begin{bmatrix} 1 & x/n \\ -x/n & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & x/n \\ -x/n & 1 \end{bmatrix} \begin{bmatrix} 1 & x/n \\ -x/n & 1 \end{bmatrix} = \begin{bmatrix} 1 - \frac{x^2}{n^2} & \frac{2x}{n} \\ -\frac{2x}{n} & 1 - \frac{x^2}{n^2} \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 - \frac{x^2}{n^2} & \frac{2x}{n} \\ -\frac{2x}{n} & 1 - \frac{x^2}{n^2} \end{bmatrix} \begin{bmatrix} 1 & x/n \\ -x/n & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 - \frac{3x^2}{n^2} & \frac{3x}{n} - \frac{x^3}{n^3} \\ -\frac{3x}{n} + \frac{x^3}{n^3} & 1 - \frac{3x^2}{n^2} \end{bmatrix}$$

$$= \begin{bmatrix} {}^3C_0 - {}^3C_2 \frac{x^2}{n^2} & {}^3C_1 \frac{x}{n} - {}^3C_3 \frac{x^3}{n^3} \\ {}^3C_1 \frac{x}{n} + {}^3C_3 \frac{x^3}{n^3} & {}^3C_0 - {}^3C_2 \frac{x^2}{n^2} \end{bmatrix}$$

⋮

$$A^n = \begin{bmatrix} {}^nC_0 - {}^nC_2 \frac{x^2}{n^2} + {}^nC_4 \frac{x^4}{n^4} \dots - {}^nC_1 \frac{x}{n} + {}^nC_3 \frac{x^3}{n^3} - {}^nC_5 \frac{x^5}{n^5} \dots \\ - {}^nC_1 \frac{x}{n} + {}^nC_3 \frac{x^3}{n^3} + {}^nC_5 \frac{x^5}{n^5} \dots - {}^nC_0 + {}^nC_2 \frac{x^2}{n^2} - {}^nC_4 \frac{x^4}{n^4} \dots \end{bmatrix}$$

$$\lim_{n \rightarrow \infty} A^n = \begin{bmatrix} 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots & x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \\ -x + \frac{x^3}{3!} - \frac{x^5}{5!} + \dots & 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \end{bmatrix}$$

$$= \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$

30. If A is a 2×2 matrix and $A^2 = I$ where $A \neq I$, $A \neq -I$ then which of the following is necessarily true?

- (1) $\text{Tr}(A) = 0$ and $|A| = 1$ or -1 (2) $\text{Tr}(A) = 0$ and $|A| = 1$
 (3) $\text{Tr}(A) \neq 0$ and $|A| = -1$ (4) $\text{Tr}(A) = 0$ and $|A| = -1$

Sol. Answer (4)

The characteristic equation of A will be $A^2 - \text{Tr}(A)A + |A|I = 0$

Comparing with $A^2 = I$

$\text{Tr}(A) = 0$ and $|A| = -1$

31. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $P = \begin{bmatrix} \cos \pi/6 & \sin \pi/6 \\ -\sin \pi/6 & \cos \pi/6 \end{bmatrix}$ and $Q = PAP^T$, then $P^T Q^{2013} P$

- (1) $\begin{bmatrix} 1 & 2013 \\ 0 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 0 & 2013 \\ 0 & 1 \end{bmatrix}$ (3) $\begin{bmatrix} 2013 & 0 \\ 0 & 2013 \end{bmatrix}$ (4) $\begin{bmatrix} 0 & 2013 \\ 2013 & 0 \end{bmatrix}$

Sol. Answer (1)

$$P^T P = I$$

$$P^T Q^{2013} P$$

$$\Rightarrow P^T Q^{2012} P A P^T P$$

$$\Rightarrow P^T Q^{2012} P A$$

$$\Rightarrow P^T Q^{2011} P A P^T P A$$

$$\Rightarrow P^T Q^{2011} P A^2$$

⋮

$$\Rightarrow P^T Q P A^{2012}$$

$$\Rightarrow P^T P A P^T P A^{2012}$$

$$\Rightarrow A^{2013} = \begin{bmatrix} 1 & 2013 \\ 0 & 1 \end{bmatrix}$$

32. Let t be the trace of the matrix $A = \begin{bmatrix} \frac{|x+y|}{|x|+|y|} & \alpha_1 & \beta_1 \\ \alpha_2 & \frac{|y+z|}{|y|+|z|} & \beta_2 \\ \alpha_3 & \beta_3 & \frac{|z+x|}{|z|+|x|} \end{bmatrix}$, then

(1) $0 \leq t \leq 3$

(2) $1 \leq t \leq 2$

(3) $1 \leq t \leq 3$

(4) $-1 \leq t \leq 1$

Sol. Answer (3)

Now, $|x+y| \leq |x|+|y|$

$$\Rightarrow \frac{|x+y|}{|x|+|y|} \leq 1$$

Hence, $t \leq 3$

Also, among the diagonal elements, at least one of the element must be 1.

33. Let A be an involutory matrix and S be the set containing solutions of $A^x = I$. Then minimum value of

$$\sum_{x \in S} (\sin^x \theta + \cos^x \theta)$$

(1) 2

(2) 4

(3) 0

(4) -2

Sol. Answer (1)

$$S = \{2, 4, 6, 8, \dots\}$$

$$(\sin^2 \theta + \cos^2 \theta) + (\sin^4 \theta + \cos^4 \theta) + (\sin^6 \theta + \cos^6 \theta) + \dots \infty$$

$$\Rightarrow \frac{\sin^2 \theta}{1 - \sin^2 \theta} + \frac{\cos^2 \theta}{1 - \cos^2 \theta} = \tan^2 \theta + \cot^2 \theta \geq 2$$

34. If A is a square matrix, then A is symmetric, iff

- (1) $A^2 = A$ (2) $A^2 = I$ (3) $A^T = A$ (4) $A^T = -A$

Sol. Answer (3)

35. If A is a square matrix, then A is skew symmetric iff

- (1) $A^2 = A$ (2) $A^2 = I$ (3) $A^T = A$ (4) $A^T = -A$

Sol. Answer (4)

36. If A is any square matrix, then

- (1) $A + A^T$ is skew symmetric (2) $A - A^T$ is symmetric
(3) AA^T is symmetric (4) AA^T is skew symmetric

Sol. Answer (3)

37. If A and B are symmetric matrices of the same order, then

- (1) AB is a symmetric matrix (2) $A - B$ is a skew-symmetric matrix
(3) $AB + BA$ is a symmetric matrix (4) $AB - BA$ is a symmetric matrix

Sol. Answer (3)

38. If A is any matrix then which of the following is not symmetric?

- (1) $A + A^T$ (2) $A - A^T$ (3) AA^T (4) $A^T A$

Sol. Answer (2)

39. Each diagonal element of a skew symmetric matrix is

- (1) Zero (2) Positive and equal
(3) Negative and equal (4) Any real number

Sol. Answer (1)

40. Let A be the set of all 3×3 matrices which are symmetric with entries 0 or 1. If there are five 1's and four 0's, then number of matrices in A

- (1) 6 (2) 12 (3) 9 (4) 18

Sol. Answer (2)

$$\begin{bmatrix} 1 & & \\ & 0 & \\ & & 0 \end{bmatrix} \text{ No. of matrices} = \frac{3!}{2!} = 3$$

$$\begin{bmatrix} 0 & & \\ & 1 & \\ & & 0 \end{bmatrix} \text{ No. of matrices} = \frac{3!}{2!} = 3$$

$$\begin{bmatrix} 0 & & \\ & 0 & \\ & & 1 \end{bmatrix} \text{ No. of matrices} = \frac{3!}{2!} = 3$$

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \text{ No. of matrices} = \frac{3!}{2!} = 3$$

41. Multiplicative inverse of the matrix $\begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix}$ is

- (1) $\begin{bmatrix} 4 & -1 \\ -7 & -2 \end{bmatrix}$ (2) $\begin{bmatrix} -4 & -1 \\ 7 & -2 \end{bmatrix}$ (3) $\begin{bmatrix} 4 & -1 \\ 7 & 2 \end{bmatrix}$ (4) $\begin{bmatrix} 4 & -1 \\ -7 & 2 \end{bmatrix}$

Sol. Answer (4)

42. The matrix X in the equation $AX = B$, such that $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ is given by

- (1) $\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 0 & -1 \\ -3 & 1 \end{bmatrix}$

Sol. Answer (2)

43. The matrix $A(I + A)^{-1}$ is equal to (if $|A| \neq 0$, $|I + A| \neq 0$)

- (1) $(A^{-1} + I)^{-1}$ (2) $A^{-1} + I$ (3) $(I + A)A$ (4) $(A^{-1} + I)^{-1}A$

Sol. Answer (1)

$$\text{Let } B = A(I + A)^{-1}$$

$$\text{Then, } B^{-1} = (I + A)A^{-1} = A^{-1} + I$$

$$B = (A^{-1} + I)^{-1}$$

44. If P is a 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3×3 identity matrix, then

there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that

[IIT-JEE 2012]

- (1) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (2) $PX = X$ (3) $PX = 2X$ (4) $PX = -X$

Sol. Answer (4)

$$P^T = 2P + I$$

$$\Rightarrow P = 2P^T + I$$

$$\Rightarrow P = 2[2P + I] + I$$

$$\Rightarrow P = 4P + 3I$$

$$\Rightarrow 3P = -3I$$

$$\Rightarrow PX = -X$$

45. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then

$$\frac{q_{31} + q_{32}}{q_{21}} \text{ equals}$$

[JEE(Advanced)-2016]

- (1) 52 (2) 103 (3) 201 (4) 205

Sol. Answer (2)

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 8 & 1 & 0 \\ 48 & 8 & 1 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 96 & 12 & 1 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 1 & 0 & 0 \\ 16 & 1 & 0 \\ 160 & 16 & 1 \end{bmatrix}$$

For the sequence 16, 48, 96, 160, ..., n^{th} term is $16 \times \frac{n(n+1)}{2}$

$$\text{So, } P^{50} = \begin{bmatrix} 1 & 0 & 0 \\ 200 & 1 & 0 \\ \frac{50 \times 51}{2} \times 16 & 200 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 200 & 1 & 0 \\ 20400 & 200 & 1 \end{bmatrix}$$

$$\text{So, } Q = P^{50} - I = \begin{bmatrix} 0 & 0 & 0 \\ 200 & 0 & 0 \\ 20400 & 200 & 0 \end{bmatrix}$$

$$\frac{q_{31} + q_{32}}{q_{21}} = \frac{20400 + 200}{200} = 103$$

46. How many 3×3 matrices M with entries from $\{0, 1, 2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5? [JEE(Advanced)-2017]

- (1) 162 (2) 135 (3) 126 (4) 198

Sol. Answer (4)

$$\text{Let } M = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, M^T = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$\text{Sum of diagonal entries} = \sum_{i=1}^3 (a_i^2 + b_i^2 + c_i^2) = 5$$

$\therefore$ Possible cases are (one 2, one 1 and seven zeros) or (five 1's and four 0's)

$$= \frac{9!}{7!} + \frac{9!}{5!4!} = 72 + 126 = 198$$

SECTION - B

Objective Type Questions (More than one options are correct)

1. Let $A(\theta) = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$ and $C(\theta, \phi) = A(\theta)A(\phi)$, then which of the following matrices is a null matrix?

(1) $C\left(\frac{2\pi}{5}, -\frac{11\pi}{10}\right)$ (2) $C\left(-\frac{11\pi}{10}, \frac{2\pi}{5}\right)$ (3) $C\left(\frac{3\pi}{7}, \frac{41\pi}{14}\right)$ (4) $C\left(\frac{7\pi}{3}, \frac{29\pi}{6}\right)$

Sol. Answer (1, 2, 3, 4)

$$C(\theta, \phi) = A(\theta)A(\phi)$$

$$\begin{aligned} &= \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta \cos^2 \phi + \cos \theta \cos \phi \sin \theta \sin \phi & \cos^2 \theta \cos \phi \sin \phi + \cos \theta \sin \theta \sin^2 \phi \\ \cos^2 \phi \cos \theta \sin \theta + \cos \phi \sin \phi \sin^2 \theta & \cos \theta \cos \phi \sin \theta \sin \phi + \sin^2 \theta \sin^2 \phi \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos \phi \cos(\theta - \phi) & \cos \theta \sin \phi \cos(\theta - \phi) \\ \sin \theta \cos \phi \cos(\theta - \phi) & \sin \theta \sin \phi \cos(\theta - \phi) \end{bmatrix} \end{aligned}$$

All options are correct, since

$$\cos\left(\frac{2\pi}{5} - \frac{11\pi}{10}\right) = \cos\left(\frac{2\pi}{5} + \frac{11\pi}{10}\right) = \cos\frac{3\pi}{2} = 0$$

$$\cos\left(\frac{-11\pi}{10} - \frac{2\pi}{5}\right) = \cos\left(\frac{-3\pi}{2}\right) = 0$$

$$\cos\left(\frac{3\pi}{7} - \frac{41\pi}{14}\right) = \cos\left(\frac{-35\pi}{14}\right) = 0$$

$$\cos\left(\frac{7\pi}{3} - \frac{29\pi}{6}\right) = \cos\left(\frac{-15\pi}{6}\right) = 0$$

2. If A and B are commuting square matrices of the same order, then which of the following is/are correct?

- (1) A and B^n commute, $n \in \mathbb{N}$ (2) A^n and B commute, $n \in \mathbb{N}$
 (3) $A - \lambda I$ and $B + \mu I$ commute, $\lambda, \mu \in \mathbb{R}$ (4) $A + \lambda I$ and $\mu I - B$ commute, $\lambda, \mu \in \mathbb{R}$

Sol. Answer (1, 2, 3, 4)

$$\begin{aligned} AB^n &= (AB)B^{n-1} \\ &= (BA)B^{n-1} \quad (\because AB = BA) \\ &= B(AB)B^{n-2} \\ &= B(BA)B^{n-2} \\ &= B^2AB^{n-2} \\ &= B^2(AB)B^{n-3} \end{aligned}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

$$= B^n A$$

$\therefore A$ and B^n commute

Similarly, A^n and B commute.

$$(A - \lambda I)(B + \mu I) = AB + \mu A I - \lambda B I + \lambda \mu I$$

$$= BA + \mu A I - \lambda B I + \lambda \mu I$$

$$= (B + \mu I)(A - \lambda I)$$

$\therefore A - \lambda I$ and $B + \mu I$ commute.

Similarly, $A + \lambda I$ and $\mu I - B$ also commute.

3. A matrix $A = [a_{ij}]_{m \times n}$ is

(1) Horizontal matrix if $m > n$

(2) Horizontal matrix if $m < n$

(3) Vertical matrix if $m > n$

(4) Vertical matrix if $m < n$

Sol. Answer (2, 3)

If $m > n \Rightarrow$ shape $\rightarrow$ vertical

If $m < n \Rightarrow$ shape $\rightarrow$ horizontal

4. If a square matrix $A = [a_{ij}]$, $a_{ij} = i^2 - j^2$ is of even order, then

(1) A is skew-symmetric

(2) $|A|$ is perfect square

(3) A is symmetric and $|A| = 0$

(4) A is neither symmetric nor skew-symmetric

Sol. Answer (1, 2)

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}$$

$$A' = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} = -A \Rightarrow A' = -A$$

$$\text{and } |A| = \begin{vmatrix} 0 & -3 \\ 3 & 0 \end{vmatrix} = 9, \text{ a perfect square}$$

5. If A and B are two square matrices such that they commute, then which of the following is true?

$$(1) AB^{2013} = B^{2013}A$$

$$(2) (AB)^{2013} = A^{2013}B^{2013}$$

$$(3) (A + B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1}B + {}^nC_2 A^{n-2}B^2 + \dots + {}^nC_n B^n$$

$$(4) A^2 - B^2 = (A - B)(A + B)$$

Sol. Answer (1, 2, 3, 4)

If A and B commute,

$$\text{then } AB^n = B^n A, (AB)^n = A^n B^n \text{ and } (A + B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1}B + {}^nC_2 A^{n-2}B^2 + \dots + {}^nC_n B^n$$

This can be proved by the principle of mathematical induction.

6. If A is a nilpotent matrix of odd order r , then which of the following is true?

- (1) $I = (I - A)(I^{r-1} + I^{r-2}A + I^{r-3}A^2 + \dots + A^{r-1})$ (2) $I = (I + A)(I^{r-1} + I^{r-2}A + I^{r-3}A^2 + \dots + A^{r-1})$
 (3) $I = (I + A)(I^{r-1} - I^{r-2}A + I^{r-3}A^2 - \dots + A^{r-1})$ (4) $I = (I - A)(I^{r-1} - I^{r-2}A + I^{r-3}A^2 - \dots + A^{r-1})$

Sol. Answer (1, 3)

Now, $A^r = 0$... (i)

Apply,

$$I^r - A^r = (I - A)(I^{r-1} + I^{r-2}A + I^{r-3}A^2 + \dots + A^{r-1})$$

$$I^r + A^r = (I + A)(I^{r-1} - I^{r-2}A + I^{r-3}A^2 - \dots + A^{r-1})$$

7. If A is any square matrix such that $A + \frac{I}{2}$ and $A - \frac{I}{2}$ are orthogonal matrices, then

- (1) A is symmetric (2) A is skew-symmetric
 (3) $A^2 = \frac{3I}{4}$ (4) $A^2 = \frac{-3I}{4}$

Sol. Answer (2, 4)

$$\left(A + \frac{I}{2}\right)\left(A^T + \frac{I}{2}\right) = I$$

$$\Rightarrow AA^T + \frac{A}{2} + \frac{A^T}{2} + \frac{I}{4} = I \quad \dots (i)$$

$$\text{Similarly, } AA^T - \frac{A}{2} - \frac{A^T}{2} + \frac{I}{4} = I \quad \dots (ii)$$

$$(i) - (ii)$$

$$A^T = -A$$

$$(i) + (ii)$$

$$AA^T = \frac{3I}{2}$$

$$\text{or } A^2 = -\frac{3I}{2}$$

8. Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when $n =$ [JEE(Advanced) 2013]

- (1) 57 (2) 55 (3) 58 (4) 56

Sol. Answer (2, 3, 4)

$$P = [P_{ij}]_{n \times n} \quad P^2 \neq 0$$

$$P_{ij} = \omega^{i+j}$$

$$P = \begin{bmatrix} \omega^2 & 1 & \omega & \omega^2 & 1 & \dots \\ 1 & \omega & \omega^2 & 1 & \omega & \dots \\ \omega & \omega^2 & 1 & \omega & \omega^2 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix}_{n \times n}$$

$$P^2 = \begin{bmatrix} \omega^2 & 1 & \omega & \omega^2 & 1 & \dots \\ 1 & \omega & \omega^2 & 1 & \omega & \dots \\ \omega & \omega^2 & 1 & \omega & \omega^2 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix} \begin{bmatrix} \omega^2 & 1 & \omega & \omega^2 & 1 & \dots \\ 1 & \omega & \omega^2 & 1 & \omega & \dots \\ \omega & \omega^2 & 1 & \omega & \omega^2 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

$$(\omega^4 + 1 + \omega^2) + (\omega^4 + 1 + \omega^2) + \dots = 0$$

Only when n is a multiple of 3.

$$\therefore n \text{ can be } 55, 58, 56 \quad (P^2 \neq 0)$$

9. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if
- (1) The first column of M is the transpose of the second row of M
 - (2) The second row of M is the transpose of the first column of M
 - (3) M is a diagonal matrix with nonzero entries in the main diagonal
 - (4) The product of entries in the main diagonal of M is not the square of an integer

[JEE(Advanced)-2014]

Sol. Answer (3, 4)

$$\text{Let } M = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \text{ where } a, b, c \in I$$

$$\text{for invertible matrix, } \det(M) \neq 0 \Rightarrow ac - b^2 \neq 0$$

$$\text{i.e. } ac \neq b^2$$

So, options (3) & (4) satisfies the above condition.

10. Let X and Y be two arbitrary, 3×3 , non-zero, skew-symmetric matrices and Z be an arbitrary 3×3 , non-zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric? [JEE(Advanced)-2015]

$$(1) Y^3 Z^4 - Z^4 Y^3 \quad (2) X^{44} + Y^{44} \quad (3) X^4 Z^3 - Z^3 X^4 \quad (4) X^{23} + Y^{23}$$

Sol. Answer (3, 4)

$$X^T = -X, Y^T = -Y \text{ and } Z^T = Z$$

$$P = Y^3 Z^4 - Z^4 Y^3$$

$$\begin{aligned} P^T &= (Z^4)^T (Y^3)^T - (Y^3)^T (Z^4)^T \\ &= (Z^T)^4 (Y^T)^3 - (Y^T)^3 (Z^T)^4 \\ &= Z^4 (-Y^3) - (-Y^3) Z^4 \\ &= Y^3 Z^4 - Z^4 Y^3 = P(\text{symmetric}) \end{aligned}$$

$$Q = X^{44} + Y^{44}$$

$$Q^T = (X^T)^{44} + (Y^T)^{44} = X^{44} + Y^{44} = Q(\text{symmetric})$$

$$R = X^4 Z^3 - Z^3 X^4$$

$$\begin{aligned} R^T &= (Z^3)^T (X^4)^T - (X^4)^T (Z^3)^T \\ &= (Z^T)^3 (X^T)^4 - (X^T)^4 (Z^T)^3 \\ &= Z^3 (-X)^4 - (-X)^4 (Z)^3 \\ &= Z^3 X^4 - X^4 Z^3 = -R(\text{Skew Symmetric}) \end{aligned}$$

$$S = X^{23} + Y^{23}$$

$$\begin{aligned} S^T &= (X^T)^{23} + (Y^T)^{23} = (-X)^{23} + (-Y)^{23} \\ &= -X^{23} - Y^{23} = -S(\text{Skew symmetric}) \end{aligned}$$

11. Which of the following is(are) not the square of a 3×3 matrix with real entries?

[JEE(Advanced)-2017]

$$(1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$(2) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$(3) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(4) \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Sol. Answer (2, 4)

Since determinant value of matrix in option (1) & (3) are positive.

Hence it can be represented as square of a matrix, but determinant value of matrix in option (2) & (4) are negative hence, it cannot be represented as square of a matrix.

12. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in \mathbb{R}$ and the system of equations (in real variables)

$$-x + 2y + 5z = b_1$$

$$2x - 4y + 3z = b_2$$

$$x - 2y + 2z = b_3$$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at least one

solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

[JEE(Advanced)-2018]

$$(1) x + 2y + 3z = b_1, 4y + 5z = b_2 \text{ and } x + 2y + 6z = b_3$$

$$(2) x + y + 3z = b_1, 5x + 2y + 6z = b_2 \text{ and } -2x - y - 3z = b_3$$

$$(3) -x + 2y - 5z = b_1, 2x - 4y + 10z = b_2 \text{ and } x - 2y + 5z = b_3$$

$$(4) x + 2y + 5z = b_1, 2x + 3z = b_2 \text{ and } x + 4y - 5z = b_3$$

Sol. Answer (1, 4)

Given system of equations can be written as

$$\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & 3 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow AX = B$$

$$[A|B] = \left[\begin{array}{ccc|c} -1 & 2 & 5 & b_1 \\ 2 & -4 & 3 & b_2 \\ 1 & -2 & 2 & b_3 \end{array} \right] \sim \left[\begin{array}{ccc|c} -1 & 2 & 5 & b_1 \\ 0 & 0 & 6 & b_1 + b_2 - b_3 \\ 0 & 0 & 7 & b_1 + b_3 \end{array} \right] \sim \left[\begin{array}{ccc|c} -1 & 2 & 5 & b_1 \\ 0 & 0 & 0 & \frac{1}{7}b_1 + b_2 - \frac{13}{7}b_3 \\ 0 & 0 & 7 & b_1 + b_3 \end{array} \right]$$

$$\text{For solution of this system, } \frac{b_1}{7} + b_2 - \frac{13b_3}{7} = 0$$

$$\Rightarrow b_1 + 7b_2 - 13b_3 = 0 \Rightarrow (b_1, b_2, b_3) = (-7K_2 + 13K_3, K_2, K_3) \text{ where } K_2, K_3 \in \mathbb{R} \quad \dots(i)$$

(A) $\Delta \neq 0 \Rightarrow$ any possible set of (b_1, b_2, b_3) would give a solution.

$\Rightarrow$ every set of values of (i) provides atleast a solution of system of option (A).

$$(B) \left[\begin{array}{ccc|c} 1 & 1 & 3 & b_1 \\ 5 & 2 & 6 & b_2 \\ -2 & -1 & -3 & b_3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 3 & b_1 \\ 3 & 0 & 0 & -2b_1 + b_2 \\ -1 & 0 & 0 & b_1 + b_3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 3 & b_1 \\ 0 & 0 & 0 & b_1 + b_2 + 3b_3 \\ -1 & 0 & 0 & b_1 + b_3 \end{array} \right]$$

For solution of this system, $b_1 + b_2 + 3b_3 = 0$

$$\Rightarrow (b_1, b_2, b_3) = (-K_2 - 3K_3, K_2, K_3), K_2, K_3 \in \mathbb{R} \quad \dots(ii)$$

Obviously, the set represented by (i) is not contained in (ii).

$\Rightarrow$ This system doesn't have solution for every set of (b_1, b_2, b_3) of (i).

(C) Equations are of parallel planes or identical planes. For solution, these planes should be identical for that, $(b_1, b_2, b_3) = (-K_3, 2K_3, K_3) \quad \dots(iii)$

Every such value of (b_1, b_2, b_3) is contained in (i) but not every value of (b_1, b_2, b_3) of (i) is present in (iii).

$\Rightarrow$ Not every (b_1, b_2, b_3) of (i) gives solution to the system of equations of option (C).

(D) $\Delta \neq 0 \Rightarrow$ every set of values of (i) provides atleast a solution of system of option (D).

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Let ψ_A be defined as trace of a matrix which is sum of diagonal elements of a square matrix.

1. $\psi_{\lambda A + \mu B} =$

where λ, μ are scalars

(1) $\lambda\psi_A + \mu\psi_B$

(2) $\lambda\psi_B + \mu\psi_A$

(3) $\lambda\psi_{AB} + \mu\psi_{BA}$

(4) None of these

Sol. Answer (1)

$$\psi_{\lambda A + \mu B} = \text{tr}(\lambda A + \mu B) = \lambda \text{tr}(A) + \mu \text{tr}(B) = \lambda\psi_A + \mu\psi_B$$

2. Which of the following are true?

(1) $\psi_{A+B} = \psi_{A-B}$

(2) $\psi_{A+B} = \psi_{AB}$

(3) $\psi_{AB} = \psi_{BA}$

(4) $\psi_{A-B} = \psi_{BA}$

Sol. Answer (3)

From theory of matrices, $\text{tr}(AB) = \text{tr}(BA)$

3. Which of the following is false?

(1) $\psi_{ABC} = \psi_{BAC}$

(2) $\psi_{ABC} = \psi_{CBA}$

(3) $\psi_{ABC} = \psi_{BCA}$

(4) None of these

Sol. Answer (4)

$$\psi_{ABC} = \text{tr}(ABC) = \text{tr}(BCA) = \psi_{BCA}$$

Comprehension-II

Let p be an odd prime number and T_p be the following set of 2×2 matrices

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, \dots, p-1\} \right\}$$

[IIT-JEE 2010]

1. The number of A in T_p such that A is either symmetric or skew-symmetric or both, and $\det(A)$ divisible by p is

(1) $(p-1)^2$

(2) $2(p-1)$

(3) $(p-1)^2 + 1$

(4) $2p-1$

Sol. Answer (4)

If A is skew-symmetric, $|A| = -b^2$

$\therefore p \mid |A|$ iff $b = 0$ in which case A is symmetric.

If A is $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$, $|A| = a^2 - b^2$

$\therefore p \mid |A|$ if either $p \mid a + b$ or $p \mid a - b$.

The second case occurs only when $a = b$ which gives p choices. The first case again gives p choices with $a = 0 = b$ counted again.

$\therefore$ The number of such matrices is $2p - 1$.

2. The number of A in T_p such that the trace of A is not divisible by p but $\det(A)$ is divisible by p is

[Note : The trace of a matrix is the sum of its diagonal entries]

(1) $(p-1)(p^2-p+1)$ (2) $p^3-(p-1)^2$ (3) $(p-1)^2$ (4) $(p-1)(p^2-2)$

Sol. Answer (3)

$\text{Tr}(A) = 2a$ will be divisible by p if and only if $a = 0$.

We want to find all matrices $\begin{bmatrix} a & b \\ c & a \end{bmatrix}$ with $a = 1, 2, \dots, p-1$ for which $a^2 - bc \equiv 0 \pmod{p}$.

There are exactly $p-1$ ordered pairs (b, c) for any value of a .

$\therefore$ Required number is $(p-1)^2$

3. The number of A in T_p such that $\det(A)$ is not divisible by p is

(1) $2p^2$ (2) p^3-5p (3) p^3-3p (4) p^3-p^2

Sol. Answer (4)

The number of matrices for which p does not divide $\text{Tr}(A)$ is $(p-1)p^2$ of these $(p-1)p^2$ are such that $p \mid |A|$.
The number of matrices for which p divides $\text{Tr}(A)$ and p does not divide $|A|$ are $(p-1)^2$.

$\therefore$ Required number is $(p-1)p^2 - (p-1)^2 + (p-1)^2 = p^3 - p^2$

SECTION - D

Assertion-Reason Type Questions

1. Let A and B be n -rowed square matrices

STATEMENT-1 : The identity $(x+y)^2 = x^2 + 2xy + y^2$ doesn't hold when x and y are substituted by A and B .

and

STATEMENT-2 : Matrix multiplication is not commutative.

Sol. Answer (1)

Statement 1 is true, since $AB \neq BA$ (Matrix multiplication is not commutative).

Hence statement 2 is a correct explanation

2. Consider the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$

STATEMENT-1 : A is idempotent.

and

STATEMENT-2 : If $A^2 = A$, then A is called idempotent.

Sol. Answer (1)

$$\begin{aligned}
 A^2 &= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \\
 &= \begin{bmatrix} 4+2-4 & -4-6+8 & -8-8+12 \\ -2-3+4 & 2+5-6 & 4+12-12 \\ 2+2-3 & -2-6+6 & -4-8+9 \end{bmatrix} \\
 &= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \\
 &= A
 \end{aligned}$$

3. STATEMENT-1 : $A = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$ is an orthogonal matrix.

and

STATEMENT-2 : If A and B are orthogonal, then AB is also orthogonal.

Sol. Answer (2)

$$\because AA^T = I \Rightarrow A \text{ is orthogonal}$$

Also, if A and B are orthogonal, then AB is orthogonal.

SECTION - E

Matrix-Match Type Questions

1. Match the following.

Column-I

- (A) If A and B are orthogonal, then AB is
 (B) If A and B are nilpotent matrices of order r and s and A and B commute, then $(AB)^r$ is
 (C) If A is a hermitian matrix such that $A^2 = 0$, then A is
 (D) If A and B are unitary matrices, then AB is

Column-II

- (p) Zero matrix
 (q) Nilpotent matrix
 (r) Unitary matrix
 (s) Orthogonal

Sol. Answer A(s), B(p, q), C(p, q), D(r)

$$(A) \quad AA^T = A^T A = I, \quad BB^T = B^T B = I$$

$$(AB)^T AB = B^T A^T AB = I$$

$$(AB)(AB)^T = I$$

$$(B) \quad A^r = 0, B^s = 0$$

$$\begin{aligned}
 (AB)^r &= A^r B^r \text{ if } A \text{ and } B \text{ commute} \\
 &= 0
 \end{aligned}$$

$$(C) A^2 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \begin{bmatrix} \bar{a}_{11} & a_{12} & a_{13} \\ \bar{a}_{12} & \bar{a}_{22} & a_{23} \\ \bar{a}_{13} & \bar{a}_{23} & \bar{a}_{33} \end{bmatrix}$$

$$= \begin{bmatrix} |a_{11}|^2 + |a_{12}|^2 + |a_{13}|^2 & a_{11}\bar{a}_{12} + a_{12}\bar{a}_{22} + a_{13}\bar{a}_{23} & a_{11}\bar{a}_{13} + a_{12}\bar{a}_{23} + a_{13}\bar{a}_{33} \\ \bar{a}_{12}\bar{a}_{11} + a_{22}\bar{a}_{12} + a_{23}\bar{a}_{13} & |a_{12}|^2 + |a_{22}|^2 + |a_{23}|^2 & \bar{a}_{12}\bar{a}_{13} + a_{22}\bar{a}_{23} + a_{23}\bar{a}_{33} \\ \bar{a}_{13}\bar{a}_{11} + \bar{a}_{23}\bar{a}_{12} + a_{33}\bar{a}_{13} & \bar{a}_{13}\bar{a}_{12} + a_{23}\bar{a}_{22} + a_{33}\bar{a}_{32} & |a_{13}|^2 + |a_{23}|^2 + |a_{33}|^2 \end{bmatrix}$$

$$\Rightarrow a_{11} = a_{12} = a_{13} = a_{21} = a_{22} = a_{23} = a_{31} = a_{32} = a_{33} = 0$$

$\Rightarrow A$ is null matrix.

$$(D) AA^0 = A^0A = I, B^0B = BB^0 = I$$

$$\text{Now, } (AB)^0 AB = B^0 A^0 AB = I$$

$$(AB)(AB)^0 = ABB^0A^0 = I$$

2. Match the Statements/Expressions in **Column-I** with the Statements/Expressions in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS. **[IIT-JEE 2008]**

Column-I**Column-II**

(A) If A is a skew-symmetric, then trace of A is (p) 0

(B) Let A and B be 3×3 matrices of real number, (q) 1

where A is symmetric, B is skew-symmetric, and

$$(A + B)(A - B) = (A - B)(A + B). \text{ If } (AB)^t = (-1)^k AB,$$

where $(AB)^t$ is the transpose of the matrix AB ,

then the possible values of k are

(C) If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 16 & 0 \\ 5 & 1 \end{bmatrix}$ and $A^2 = B$, then the (r) 2

value of α is

(D) Given the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, the constants (s) 3

p, q, r such that $A^3 = pA^2 + qA + rI$, then the value of $p + q + r$ is

(t) 4

Sol. Answer A(p), B(q, s), C(t), D(r)

(A) As all the elements of principal diagonal of skew symmetric matrix are zero, hence the trace will be equal to 0.

$$(B) (A + B)(A - B) = (A - B)(A + B)$$

$$\Rightarrow A^2 - AB + BA - B^2 = A^2 + AB - BA - B^2$$

$$\Rightarrow AB = BA$$

$$(AB)^t = B^t A^t$$

$$= -BA$$

$$= -AB$$

$$\therefore k = \text{odd}$$

(C) As $A^2 = B$

$$\Rightarrow \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 5 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 5 & 1 \end{bmatrix}$$

$$\Rightarrow \alpha^2 = 16 \text{ and } \alpha + 1 = 5$$

$$\Rightarrow \alpha = \pm 4 \text{ and } \alpha = 4$$

$\Rightarrow$ Taking intersection, we will get $\alpha = 4$

(D) $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & 2 & -1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 + \lambda^2 - 2\lambda - 1 = 0$$

$$\Rightarrow A^3 + A^2 - 2A - 1 = 0 \quad \dots(i)$$

$$\text{Given that, } A^3 = pA^2 + qA + rD \quad \dots(ii)$$

On comparing (i) and (ii), we get

$$p = -1, q = 2, r = 1$$

$$\therefore p + q + r = -1 + 2 + 1 = 2$$

SECTION - F

Integer Answer Type Questions

1. Let $A = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ and $(A + I)^{100} - 100A = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$, then $\alpha + \beta + \gamma + \delta = \dots$

Sol. Answer (2)

$$A^2 = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(A + I)^{100} - 100A = {}^{100}C_0 I^{100} = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \alpha + \beta + \gamma + \delta = 2$$

2. Let $A = \begin{bmatrix} \omega & -\omega \\ -\omega & \omega \end{bmatrix}$, where ω is a complex cube root of unity, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ and $A^9 = 2^k B$, where $k = \dots$

Sol. Answer (8)

$$A = \omega B$$

$$A^3 = B^3$$

$$\text{Now, } B^2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 2B$$

$$B^3 = B^2 \cdot B = 2B \cdot B = 2B^2 = 4B$$

$$\Rightarrow B^9 = 2^6 B^3 = 2^8 B$$

$$\text{Hence, } A^9 = B^9 = 2^8 B$$

3. If A is skew-symmetric of order 3×3 , then trace of A is

Sol. Answer (0)

$$\because A^T = -A \text{ an skew-symmetric}$$

$$\Rightarrow \text{Sum of all diagonal elements is 0.}$$

4. Let A be a square matrix of 2×2 satisfying

i. $a_{ij} = 1$ or -1

ii. $a_{11}a_{21} + a_{12}a_{22} = 0$

Then, number of matrices of A -----

Sol. Answer (8)

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}, \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

5. For a real number α , if the system $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ of linear equations, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$

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Sol. Answer (1)

Equation is rewritten as $AX = B$

$$\text{where } A = \begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow X = \frac{\text{adj}(A) \cdot B}{\det(A)}$$

For infinite solution, $\det(A) = 0$ and $\text{adj}(A) \cdot B = 0$

$$\Rightarrow \alpha = 1, -1$$

but α is not equal to 1 as in this case equation is inconsistent.

$$\text{So, } \alpha = -1$$

$$\text{i.e., } \boxed{1 + \alpha + \alpha^2 = 1}$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : All positive odd integral powers of a skew-symmetric matrix are symmetric.
 STATEMENT-2 : All positive even integral powers of a skew-symmetric matrix are symmetric.
 STATEMENT-3 : If A is a skew-symmetric matrix of even order then $|A|$ is perfect square
- (1) F T T (2) T T T (3) T F T (4) T T F

Sol. Answer (1)

Let A be skew-symmetric, the $A^T = -A$

$$\therefore (A^n)^T = (A^T)^n \quad \forall n \in N$$

$$\therefore (A^n)^T = (-A)^n$$

$$\Rightarrow (A^n)^T = \begin{cases} A^n; n \text{ is even} \\ -A^n; n \text{ is odd} \end{cases}$$

2. If A and B are symmetric matrices of same order, then

STATEMENT-1 : $A + B$ is skew-symmetric matrix.

STATEMENT-2 : $AB - BA$ is skew-symmetric matrix.

STATEMENT-3 : $A - B$ is skew-symmetric matrix.

- (1) T T T (2) F T F (3) F T T (4) F F F

Sol. Answer (2)

SECTION - H

Aakash Challengers Questions

1. If A and B are two square matrices of the order 3, then the value of $998 \operatorname{tr}(I) - 999 \operatorname{tr}(AB) + 999 \operatorname{tr}(BA)$ is

Sol. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ and $\operatorname{tr}(I) = 3$

$$\therefore \text{The required value} = 998 \times 3 = 2994$$

2. The matrix $A = \begin{bmatrix} \lambda_1^2 & \lambda_1 \lambda_2 & \lambda_1 \lambda_3 \\ \lambda_2 \lambda_1 & \lambda_2^2 & \lambda_2 \lambda_3 \\ \lambda_3 \lambda_1 & \lambda_3 \lambda_2 & \lambda_3^2 \end{bmatrix}$ is idempotent if $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 = k$ where $\lambda_1, \lambda_2, \lambda_3$ are non-zero real

numbers. Then the value of $(10 + k)^2$ is....

Sol. Answer $A^2 = A$

$$\Rightarrow \begin{bmatrix} \lambda_1^2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) & \lambda_1 \lambda_2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) & \lambda_1 \lambda_3(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \\ \lambda_2 \lambda_1(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) & \lambda_2^2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) & \lambda_2 \lambda_3(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \\ \lambda_3 \lambda_1(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) & \lambda_3 \lambda_2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) & \lambda_3^2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \end{bmatrix} = \begin{bmatrix} \lambda_1^2 & \lambda_1 \lambda_2 & \lambda_1 \lambda_3 \\ \lambda_2 \lambda_1 & \lambda_2^2 & \lambda_2 \lambda_3 \\ \lambda_3 \lambda_1 & \lambda_3 \lambda_2 & \lambda_3^2 \end{bmatrix}$$

$$\Rightarrow \lambda_1^2 + \lambda_2^2 + \lambda_3^2 = 1 = k$$

$$\therefore (10 + k)^2 = (10 + 1)^2 = 121$$

3. Find all solutions of the matrix equation $X^2 = I$, where I is the 2-rowed unit matrix, and X is a real matrix, i.e. a matrix all of whose elements are real.

Sol. Answer Let $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

$$\text{then } X^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & b(a+d) \\ c(a+d) & d^2 + bc \end{bmatrix}$$

$X^2 = I$ yields

$$\begin{bmatrix} a^2 + bc & b(a+d) \\ c(a+d) & d^2 + bc \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Equality of matrices yields

$$a^2 + bc = d^2 + bc = 1 \quad \dots(A)$$

$$b(a+d) = c(a+d) = 0 \quad \dots(B)$$

Case I

If $a + d \neq 0$ then $b = 0$, $c = 0$, from (B)

and from (A), $a^2 = d^2 = 1$

so that $a = 1$, $d = 1$ or $a = -1$, $d = -1$

Thus in this case $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ or $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ i.e., $X = \pm I$

Case II

If $a + d = 0$, i.e. $d = -a$, then

from (A)

$$a^2 + bc = d^2 + bc = 1$$

$$\Rightarrow a^2 + bc = (-a)^2 + bc = 1$$

$$\text{i.e. } a^2 + bc = 1$$

Let $a = \alpha$, $b = \beta$, $c = \gamma$, then $d = -\alpha$

Thus the matrix satisfying $X^2 = I$ can be put in the form $X = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$, where $\alpha^2 + \beta\gamma = 1$

4. The matrix $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ is

(1) Unitary

(2) Orthogonal

(3) Nilpotent

(4) Involutory

Sol. Answer (3)

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}, A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

Hence Nilpotent,

5. Let A be an idempotent matrix and $(I + A)^{100} = I + (2^{20k} - 1)A$, then $k = \dots$

Sol. Answer (5)

$$(I + A)^{100} = {}^{100}C_0 I + {}^{100}C_1 A + {}^{100}C_2 A^2 + \dots + {}^{100}C_{100} A^{100} = I + (2^{100} - 1)A$$

6. Let $A = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ such that $A^3 = 0$, then sum of all the elements of A^2 is

Sol. Answer (0)

$$\text{Now, } A^2 - (\alpha + \delta)A + (\alpha\delta - \beta\gamma)I = 0$$

Multiple by A ,

$$A^3 - (\alpha + \delta)A^2 + (\alpha\delta - \beta\gamma)A = 0 \quad \dots(i)$$

$$\text{Also, } A^3 = 0, \alpha\delta - \beta\gamma = 0$$

$$\Rightarrow -(\alpha + \delta)A^2 = 0$$

$$\Rightarrow \alpha + \delta = 0 \text{ or } A^2 = 0$$

$$\text{If } \alpha + \delta = 0, A^2 = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

$$= \begin{bmatrix} \alpha^2 + \beta\gamma & \alpha\beta + \beta\delta \\ \alpha\gamma + \gamma\delta & \delta^2 + \beta\gamma \end{bmatrix}$$

$$\text{Sum of all the elements of } A^2 = \alpha^2 + \delta^2 + 2\beta\gamma$$

$$= (\alpha + \delta)^2 = 0, \text{ because } \alpha\delta = \beta\gamma$$



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